

Integral Calculus (2005–2026 Archive)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

66

TOTAL MARKS

139 M

TEST DURATION

198 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2005–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	35	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	3	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	28	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q55 • ID: JAM_2026_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55	The volume of the tetrahedron bounded by the planes $x = 1, y = 2, z = 3$ and $12x + 8y + 6z = 70$ is _____ (rounded off to one decimal place).

Question 2 (JAM 2026 • Q53 • ID: JAM_2026_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53	The double integral of $f(x, y) = x$ over the triangular region with vertices at $\left(-\frac{1}{2}, \frac{1}{2}\right), (1, 2)$ and $(1, -1)$ is _____ (rounded off to one decimal place).
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Question 3 (JAM 2026 • Q22 • ID: JAM_2026_Q22)

MCQ

+2 M

Neg: -0.67

Q.22	Let S be the solid of intersection of two solid spheres S_1, S_2 given by $S_1: x^2 + y^2 + (z - 1)^2 \leq 4 \quad \text{and} \quad S_2: x^2 + y^2 + (z + 1)^2 \leq 4 .$ Which ONE of the following iterated integrals expresses the volume of S?
(A)	$8 \int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} \left(\sqrt{4-x^2-y^2} - 1 \right) dy dx$
(B)	$8 \int_0^2 \int_0^{\sqrt{4-x^2}} \left(\sqrt{4-x^2-y^2} - 1 \right) dy dx$
(C)	$2 \int_0^2 \int_0^{\sqrt{4-x^2}} \left(\sqrt{4-x^2-y^2} - 1 \right) dy dx$
(D)	$2 \int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} \left(\sqrt{4-x^2-y^2} - 1 \right) dy dx$

Question 4 (JAM 2026 • Q20 • ID: JAM_2026_Q20)

MCQ

+2 M

Neg: -0.67

Q.20	The semi-circle $y = \sqrt{6x - 5 - x^2}$ is rotated clockwise by the angle $\frac{\pi}{2}$ in the plane about the origin. What is the area of the planar region traced out by the semi-circle?
(A)	6π
(B)	4π
(C)	2π
(D)	π

Question 5 (JAM 2026 • Q10 • ID: JAM_2026_Q10)

MCQ

+1 M

Neg: -0.33

Q.10	Let T be the triangular region in the plane with vertices at $(0,0)$, $(1,0)$ and $(1,1)$. Let f be a function defined from T to $\mathbb{R}$ given in polar coordinates. Then the double integral of f over T is equal to _____.
(A)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr + \int_1^{\sqrt{2}} \left(\int_0^{\cos^{-1}(1/r)} f \, d\theta \right) r \, dr$
(B)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr - \int_1^{\sqrt{2}} \left(\int_0^{\cos^{-1}(1/r)} f \, d\theta \right) r \, dr$
(C)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr + \int_1^{\sqrt{2}} \left(\int_0^{\sin^{-1}(1/r)} f \, d\theta \right) r \, dr$
(D)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr - \int_1^{\sqrt{2}} \left(\int_0^{\sin^{-1}(1/r)} f \, d\theta \right) r \, dr$

Question 6 (JAM 2026 • Q5 • ID: JAM_2026_Q5)

MCQ

+1 M

Neg: -0.33

Q.5

Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function. Then

$$\int_0^2 \int_0^x \int_0^y f(x, y, z) dz dy dx = \text{_____}.$$

(A) $\int_0^2 \int_0^y \int_0^2 f(x, y, z) dx dz dy$

(B) $\int_0^2 \int_0^x \int_z^2 f(x, y, z) dy dz dx$

(C) $\int_0^2 \int_y^2 \int_0^x f(x, y, z) dz dx dy$

(D) $\int_0^2 \int_z^2 \int_0^x f(x, y, z) dy dx dz$

Question 7 (JAM 2025 • Q57 • ID: JAM_2025_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57

Let Ω be the solid bounded by the planes $z = 0$, $y = 0$, $x = \frac{1}{2}$, $2y = x$ and $2x + y + z = 4$.

If V is the volume of Ω , then the value of $64V$ is equal to _____
(rounded off to two decimal places).

Question 8 (JAM 2025 • Q48 • ID: JAM_2025_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let T denote the triangle in the xy plane bounded by the x axis and the lines $y = x$ and $x = 1$. The value of the double integral (over T)

$$\iint_T (5 - y) dx dy$$

is equal to _____ (rounded off to two decimal places).

Question 9 (JAM 2025 • Q45 • ID: JAM_2025_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45 Let S be the surface area of the portion of the plane $z = x + y + 3$, which lies inside the cylinder $x^2 + y^2 = 1$.

Then, the value of $\left(\frac{S}{\pi}\right)^2$ is equal to _____ (rounded off to two decimal places).

Question 10 (JAM 2025 • Q28 • ID: JAM_2025_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 Let Ω be the bounded region in $\mathbb{R}^3$ lying in the first octant ($x \geq 0$, $y \geq 0$, $z \geq 0$), and bounded by the surfaces $z = x^2 + y^2$, $z = 4$, $x = 0$ and $y = 0$.

Then, the volume of Ω is equal to

- (A) π
- (B) 2π
- (C) 3π
- (D) 4π

Question 11 (JAM 2025 • Q10 • ID: JAM_2025_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 The value of

$$\int_0^1 \left(\int_{\sqrt{y}}^1 3e^{x^3} dx \right) dy$$

is equal to

- (A) $e - 1$
- (B) $\frac{e-1}{2}$
- (C) $\sqrt{e} - 1$
- (D) $\frac{\sqrt{e}-1}{2}$

Question 12 (JAM 2024 • Q57 • ID: JAM_2024_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 Let $c > 0$ be such that

$$\int_0^c e^{s^2} ds = 3$$

Then, the value of

$$\int_0^c \left(\int_x^c e^{x^2+y^2} dy \right) dx$$

equals _____ (rounded off to one decimal place).

Question 13 (JAM 2024 • Q54 • ID: JAM_2024_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 Let

$$S = \{(x, y, z) \in \mathbb{R}^3: x^2 + y^2 + z^2 < 1\}.$$

Then, the value of

$$\frac{1}{\pi} \iiint_S \left((x - 2y + z)^2 + (2x - y - z)^2 + (x - y + 2z)^2 \right) dx dy dz$$

equals _____ (rounded off to two decimal places).

Question 14 (JAM 2024 • Q53 • ID: JAM_2024_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let T be the planar region enclosed by the square with vertices at the points $(0,1)$, $(1,0)$, $(0,-1)$ and $(-1,0)$. Then, the value of

$$\iint_T \left(\cos(\pi(x-y)) - \cos(\pi(x+y)) \right)^2 dx dy$$

equals _____ (rounded off to two decimal places).

Question 15 (JAM 2024 • Q48 • ID: JAM_2024_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 4, (x-1)^2 + y^2 \leq 1, z \geq 0\}.$$

Then, the surface area of S equals _____ (rounded off to two decimal places).

Question 16 (JAM 2024 • Q41 • ID: JAM_2024_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41 The area of the region

$$R = \left\{ (x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1 \text{ and } \frac{1}{4} \leq xy \leq \frac{1}{2} \right\}$$

is _____ (rounded off to two decimal places).

Question 17 (JAM 2024 • Q27 • ID: JAM_2024_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 For $a > b > 0$, consider

$$D = \left\{ (x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq a^2 \text{ and } x^2 + y^2 \geq b^2 \right\}.$$

Then, the surface area of the boundary of the solid D is

- (A) $4\pi(a+b)\sqrt{a^2-b^2}$
- (B) $4\pi(a^2-b^2)\sqrt{a^2-b^2}$
- (C) $4\pi(a-b)\sqrt{a^2-b^2}$
- (D) $4\pi(a^2+b^2)\sqrt{a^2-b^2}$

Question 18 (JAM 2023 • Q55 • ID: JAM_2023_Q55)

NAT

+2 M

Neg: Nil (0)

Q. 55 Let S be the triangular region whose vertices are $(0, 0)$, $(0, \frac{\pi}{2})$, and $(\frac{\pi}{2}, 0)$. The value of $\iint_S \sin(x) \cos(y) dx dy$ is equal to _____.
(rounded off to two decimal places)

Question 19 (JAM 2023 • Q48 • ID: JAM_2023_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let V be the volume of the region $S \subseteq \mathbb{R}^3$ defined by

$$S = \{(x, y, z) \in \mathbb{R}^3 : xy \leq z \leq 4, 0 \leq x^2 + y^2 \leq 1\}$$

Then $\frac{V}{\pi}$ is equal to _____. (rounded off to two decimal places)

Question 20 (JAM 2023 • Q31 • ID: JAM_2023_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 For each $t \in (0, 1)$, the surface P_t in $\mathbb{R}^3$ is defined by

$$P_t = \{(x, y, z) : (x^2 + y^2)z = 1, t^2 \leq x^2 + y^2 \leq 1\}.$$

Let $a_t \in \mathbb{R}$ be the surface area of P_t . Then

(A) $a_t = \iint_{t^2 \leq x^2 + y^2 \leq 1} \sqrt{1 + \frac{4x^2}{(x^2 + y^2)^4} + \frac{4y^2}{(x^2 + y^2)^4}} dx dy$

(B) $a_t = \iint_{t^2 \leq x^2 + y^2 \leq 1} \sqrt{1 + \frac{4x^2}{(x^2 + y^2)^2} + \frac{4y^2}{(x^2 + y^2)^2}} dx dy$

(C) the limit $\lim_{t \rightarrow 0^+} a_t$ does NOT exist

(D) the limit $\lim_{t \rightarrow 0^+} a_t$ exists

Question 21 (JAM 2023 • Q26 • ID: JAM_2023_Q26)

MCQ

+2 M

Neg: -0.67

Q. 26 Let

$$f(x, y) = \iint_{(u-x)^2 + (v-y)^2 \leq 1} e^{-\sqrt{(u-x)^2 + (v-y)^2}} du dv.$$

Then $\lim_{n \rightarrow \infty} f(n, n^2)$ is

- (A) non-existent
- (B) 0
- (C) $\pi(1 - e^{-1})$
- (D) $2\pi(1 - 2e^{-1})$

Question 22 (JAM 2023 • Q18 • ID: JAM_2023_Q18)

MCQ

+2 M

Neg: -0.67

Q. 18 The area of the curved surface

$$S = \{(x, y, z) \in \mathbb{R}^3 : z^2 = (x-1)^2 + (y-2)^2\}$$

lying between the planes $z = 2$ and $z = 3$ is

- (A) $4\pi\sqrt{2}$
- (B) $5\pi\sqrt{2}$
- (C) 9π
- (D) $9\pi\sqrt{2}$

Question 23 (JAM 2023 • Q7 • ID: JAM_2023_Q7)

MCQ

+1 M

Neg: -0.33

Q. 7 The value of

$$\int_0^1 \int_0^{1-x} \cos(x^3 + y^2) dy dx - \int_0^1 \int_0^{1-y} \cos(x^3 + y^2) dx dy$$

is

(A) 0

(B) $\frac{\cos(1)}{2}$

(C) $\frac{\sin(1)}{2}$

(D) $\cos\left(\frac{1}{2}\right) - \sin\left(\frac{1}{2}\right)$

Question 24 (JAM 2022 • Q52 • ID: JAM_2022_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52

If G is the region in $\mathbb{R}^2$ given by

$$G = \left\{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1, \frac{x}{\sqrt{3}} < y < \sqrt{3}x, x > 0, y > 0 \right\}$$

then the value of

$$\frac{200}{\pi} \iint_G x^2 dx dy$$

is equal to _____. (Rounded off to two decimal places)

Question 25 (JAM 2022 • Q48 • ID: JAM_2022_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48

Consider the region

$$G = \{(x, y, z) \in \mathbb{R}^3 : 0 < z < x^2 - y^2, x^2 + y^2 < 1\}.$$

Then the volume of G is equal to _____. (Rounded off to two decimal places)

Question 26 (JAM 2022 • Q11 • ID: JAM_2022_Q11)

MCQ

+2 M

Neg: -0.67

Q.11

Consider the open rectangle $G = \{(s, t) \in \mathbb{R}^2 : 0 < s < 1 \text{ and } 0 < t < 1\}$ and the map $T: G \rightarrow \mathbb{R}^2$ given by

$$T(s, t) = \left(\frac{\pi s(1-t)}{2}, \frac{\pi(1-s)}{2} \right) \quad \text{for } (s, t) \in G.$$

Then the area of the image $T(G)$ of the map T is equal to

(A)

$$\frac{\pi}{4}$$

(B)

$$\frac{\pi^2}{4}$$

(C)

$$\frac{\pi^2}{8}$$

(D)

$$1$$

Question 27 (JAM 2021 • Q43 • ID: JAM_2021_Q43)

NAT

+1 M

Neg: Nil (0)

Q. 43 Let $B = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ and define $u(x, y, z) = \sin((1 - x^2 - y^2 - z^2)^2)$ for $(x, y, z) \in B$. Then the value of

$$\iiint_B \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx dy dz$$

is _____.

Question 28 (JAM 2021 • Q5 • ID: JAM_2021_Q5)

MCQ

+1 M

Neg: -0.33

Q. 5 Let p and t be positive real numbers. Let D_t be the closed disc of radius t centered at $(0, 0)$, i.e., $D_t = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq t^2\}$. Define

$$I(p, t) = \iint_{D_t} \frac{dx dy}{(p^2 + x^2 + y^2)^p}.$$

Then $\lim_{t \rightarrow \infty} I(p, t)$ is finite

(A) only if $p > 1$.

(B) only if $p = 1$.

(C) only if $p < 1$.

(D) for no value of p .

Question 29 (JAM 2020 • Q47 • ID: JAM_2020_Q47)

NAT

+1 M

Neg: Nil (0)

Q. 47 If $\int_0^1 \int_{2y}^2 e^{x^2} dx dy = k(e^4 - 1)$, then k equals _____.

Question 30 (JAM 2020 • Q38 • ID: JAM_2020_Q38)

MSQ

+2 M

Neg: Nil (0)

Q. 38 Let S be that part of the surface of the paraboloid $z = 16 - x^2 - y^2$ which is above the plane $z = 0$ and D be its projection on the xy -plane. Then the area of S equals

(A) $\iint_D \sqrt{1 + 4(x^2 + y^2)} dx dy$

(B) $\iint_D \sqrt{1 + 2(x^2 + y^2)} dx dy$

(C) $\int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2} dr d\theta$

(D) $\int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2} r dr d\theta$

Question 31 (JAM 2020 • Q25 • ID: JAM_2020_Q25)

MCQ

+2 M

Neg: -0.67

Q. 25 Let S be the part of the cone $z^2 = x^2 + y^2$ between the planes $z = 0$ and $z = 1$. Then the value of the surface integral $\iint_S (x^2 + y^2) dS$ is

(A) π (B) $\frac{\pi}{\sqrt{2}}$ (C) $\frac{\pi}{\sqrt{3}}$ (D) $\frac{\pi}{2}$

MA

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Question 32 (JAM 2020 • Q24 • ID: JAM_2020_Q24)

MCQ

+2 M

Neg: -0.67

Q. 24 The value of the triple integral $\iiint_V (x^2y + 1) dx dy dz$, where V is the region given by $x^2 + y^2 \leq 1, 0 \leq z \leq 2$ is

(A) π (B) 2π (C) 3π (D) 4π

Question 33 (JAM 2020 • Q14 • ID: JAM_2020_Q14)

MCQ

+2 M

Neg: -0.67

Q. 14 Let $a \in \mathbb{R}$. If $f(x) = \begin{cases} (x+a)^2, & x \leq 0 \\ (x+a)^3, & x > 0, \end{cases}$

then

(A) $\frac{d^2f}{dx^2}$ does not exist at $x = 0$ for any value of a (B) $\frac{d^2f}{dx^2}$ exists at $x = 0$ for exactly one value of a (C) $\frac{d^2f}{dx^2}$ exists at $x = 0$ for exactly two values of a (D) $\frac{d^2f}{dx^2}$ exists at $x = 0$ for infinitely many values of a

$$\int x^2 \sin \frac{1}{x} + y^2 \sin \frac{1}{y}, \quad xy \neq 0$$

Question 34 (JAM 2019 • Q51 • ID: JAM_2019_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.

- Q.51 The volume of the solid of revolution of the loop of the curve $y^2 = x^4(x + 2)$ about the x -axis (round off to 2 decimal places) is _____

MA

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Question 35 (JAM 2019 • Q50 • ID: JAM_2019_Q50)

NAT

+1 M

Neg: Nil (0)

- Q.50 The volume of the solid bounded by the surfaces $x = 1 - y^2$ and $x = y^2 - 1$, and the planes $z = 0$ and $z = 2$ (round off to 2 decimal places) is _____

Q. 51 – Q. 60 carry two marks each

Question 36 (JAM 2019 • Q49 • ID: JAM_2019_Q49)

NAT

+1 M

Neg: Nil (0)

- Q.49 The value of the integral

$$\int_{-1}^1 \int_{-1}^1 |x + y| dx dy$$

(round off to 2 decimal places) is _____

Question 37 (JAM 2019 • Q26 • ID: JAM_2019_Q26)

MCQ

+2 M

Neg: -0.67

- Q.26 The tangent line to the curve of intersection of the surface $x^2 + y^2 - z = 0$ and the plane $x + z = 3$ at the point $(1, 1, 2)$ passes through

(A) $(-1, -2, 4)$ (B) $(-1, 4, 4)$ (C) $(3, 4, 4)$ (D) $(-1, 4, 0)$

Question 38 (JAM 2019 • Q10 • ID: JAM_2019_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 The area of the surface generated by rotating the curve $x = y^3$, $0 \leq y \leq 1$, about the y -axis, is

- (A) $\frac{\pi}{27} 10^{3/2}$ (B) $\frac{4\pi}{3} (10^{3/2} - 1)$ (C) $\frac{\pi}{27} (10^{3/2} - 1)$ (D) $\frac{4\pi}{3} 10^{3/2}$

Q. 11 – Q. 30 carry two marks each.

Question 39 (JAM 2018 • Q58 • ID: JAM_2018_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 The area of the parametrized surface

$$S = \{((2 + \cos u) \cos v, (2 + \cos u) \sin v, \sin u) \in \mathbb{R}^3 \mid 0 \leq u \leq \frac{\pi}{2}, 0 \leq v \leq \frac{\pi}{2}\}$$

is _____ (correct up to two decimal places).

Question 40 (JAM 2018 • Q56 • ID: JAM_2018_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 The value of the integral

$$\int_0^1 \int_x^1 y^4 e^{xy^2} dy dx$$

is _____ (correct up to three decimal places).

Question 41 (JAM 2018 • Q54 • ID: JAM_2018_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 If the volume of the solid in $\mathbb{R}^3$ bounded by the surfaces

$$x = -1, \quad x = 1, \quad y = -1, \quad y = 1, \quad z = 2, \quad y^2 + z^2 = 2$$

is $\alpha - \pi$, then $\alpha =$ _____

Question 42 (JAM 2018 • Q20 • ID: JAM_2018_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 If $\vec{F}(x, y) = (3x - 8y)\hat{i} + (4y - 6xy)\hat{j}$ for $(x, y) \in \mathbb{R}^2$, then $\oint_C \vec{F} \cdot d\vec{r}$, where C is the boundary of the triangular region bounded by the lines $x = 0, y = 0$ and $x + y = 1$ oriented in the anti-clockwise direction, is

(A) $\frac{5}{2}$

(B) 3

(C) 4

(D) 5

Question 43 (JAM 2018 • Q10 • ID: JAM_2018_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 Let $s_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}$ for $n \in \mathbb{N}$. Then which one of the following is TRUE for the sequence $\{s_n\}_{n=1}^{\infty}$

(A) $\{s_n\}_{n=1}^{\infty}$ converges in $\mathbb{Q}$ (B) $\{s_n\}_{n=1}^{\infty}$ is a Cauchy sequence but does not converge in $\mathbb{Q}$ (C) the subsequence $\{s_{k^n}\}_{n=1}^{\infty}$ is convergent in $\mathbb{R}$, only when k is even natural number(D) $\{s_n\}_{n=1}^{\infty}$ is not a Cauchy sequence

MA

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Question 44 (JAM 2017 • Q56 • ID: JAM_2017_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 For a real number x , define $\lceil x \rceil$ to be the smallest integer greater than or equal to x . Then

$$\int_0^1 \int_0^1 \int_0^1 (\lceil x \rceil + \lceil y \rceil + \lceil z \rceil) dx dy dz = \underline{\hspace{2cm}}$$

Question 45 (JAM 2017 • Q32 • ID: JAM_2017_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 The volume of the solid

$$\left\{ (x, y, z) \in \mathbb{R}^3 \mid 1 \leq x \leq 2, \quad 0 \leq y \leq \frac{2}{x}, \quad 0 \leq z \leq x \right\}$$

is expressible as

(A) $\int_1^2 \int_0^{2/x} \int_0^x dz \, dy \, dx$

(B) $\int_1^2 \int_0^x \int_0^{2/x} dy \, dz \, dx$

(C) $\int_0^2 \int_1^z \int_0^{2/x} dy \, dx \, dz$

(D) $\int_0^2 \int_{\max\{z, 1\}}^2 \int_0^{2/x} dy \, dx \, dz$

Question 46 (JAM 2017 • Q16 • ID: JAM_2017_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 The area of the surface $z = \frac{xy}{3}$ intercepted by the cylinder $x^2 + y^2 \leq 16$ lies in the interval

(A) $(20\pi, 22\pi]$

(B) $(22\pi, 24\pi]$

(C) $(24\pi, 26\pi]$

(D) $(26\pi, 28\pi]$

Question 47 (JAM 2017 • Q6 • ID: JAM_2017_Q6)

MCQ

+1 M

Neg: -0.33

Q.6

$$\int_0^1 \int_x^1 \sin(y^2) dy \, dx =$$

(A) $\frac{1+\cos 1}{2}$

(B) $1 - \cos 1$

(C) $1 + \cos 1$

(D) $\frac{1-\cos 1}{2}$

Question 48 (JAM 2016 • Q57 • ID: JAM_2016_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 The value of the double integral

$$\int_0^\pi \int_0^x \frac{\sin y}{\pi - y} dy \, dx$$

is _____

Question 49 (JAM 2016 • Q52 • ID: JAM_2016_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52 Let R be the region enclosed by $x^2 + 4y^2 \geq 1$ and $x^2 + y^2 \leq 1$. Then the value of

$$\iint_R |xy| \, dx \, dy \quad \text{is } \underline{\hspace{2cm}}$$

Question 50 (JAM 2016 • Q30 • ID: JAM_2016_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x(1 + x^\alpha \sin(\ln x^2)) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Then, at $x = 0$, the function f is

- (A) continuous and differentiable when $\alpha = 0$
- (B) continuous and differentiable when $\alpha > 0$
- (C) continuous and differentiable when $-1 < \alpha < 0$
- (D) continuous and differentiable when $\alpha < -1$

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each.

Question 51 (JAM 2016 • Q20 • ID: JAM_2016_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 Let S be the surface of the cone $z = \sqrt{x^2 + y^2}$ bounded by the planes $z = 0$ and $z = 3$. Further, let C be the closed curve forming the boundary of the surface S . A vector field $\vec{F}$ is such that $\nabla \times \vec{F} = -x\hat{i} - y\hat{j}$. The absolute value of the line integral $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$, is

- (A) 0
- (B) 9π
- (C) 15π
- (D) 18π

Question 52 (JAM 2016 • Q10 • ID: JAM_2016_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 Let S be a closed subset of $\mathbb{R}$, T a compact subset of $\mathbb{R}$ such that $S \cap T \neq \emptyset$. Then $S \cap T$ is

- (A) closed but not compact
- (B) not closed
- (C) compact
- (D) neither closed nor compact

Q. 11 – Q. 30 carry two marks each.

Question 53 (JAM 2015 • Q58 • ID: JAM_2015_Q58)

NAT

+2 M

Neg: Nil (0)

Q.18

Let ℓ be the length of the portion of the curve $x = x(y)$ between the lines $y = 1$ and $y = 3$, where $x(y)$ satisfies

$$\frac{dx}{dy} = \frac{\sqrt{1 + y^2 + y^4}}{y}, \quad x(1) = 0$$

The value of ℓ , correct upto three decimal places, is _____

Question 54 (JAM 2015 • Q55 • ID: JAM_2015_Q55)

NAT

+2 M

Neg: Nil (0)

Q.15

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{y}{\sin y}, & y \neq 0 \\ 1, & y = 0 \end{cases}$$

Then the integral

$$\frac{1}{\pi^2} \int_{x=0}^1 \int_{y=\sin^{-1} x}^{\frac{\pi}{2}} f(x, y) dy dx$$

correct upto three decimal places, is _____

Question 55 (JAM 2015 • Q42 • ID: JAM_2015_Q42)

NAT

+1 M

Neg: Nil (0)

Q.2

Let S be the portion of the surface $z = \sqrt{16 - x^2}$ bounded by the planes $x = 0$, $x = 2$, $y = 0$, and $y = 3$. The surface area of S , correct upto three decimal places, is _____

Question 56 (JAM 2015 • Q26 • ID: JAM_2015_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 The area of the planar region bounded by the curves $x = 6y^2 - 2$ and $x = 2y^2$ is

(A) $\frac{\sqrt{2}}{3}$

(B) $\frac{2\sqrt{2}}{3}$

(C) $\frac{4\sqrt{2}}{3}$

(D) $\sqrt{2}$

Question 57 (JAM 2015 • Q10 • ID: JAM_2015_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 Let A be a nonempty subset of $\mathbb{R}$. Let $I(A)$ denote the set of interior points of A . Then $I(A)$ can be

(A) empty

(B) singleton

(C) a finite set containing more than one element

(D) countable but not finite

Q. 11 – Q. 30 carry two marks each.

Question 58 (JAM 2014 • Q27 • ID: JAM_2014_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 The value of $\int_{x=0}^1 \int_{y=0}^{x^2} \int_{z=0}^y (y+2z) dz dy dx$ is

(A) $\frac{1}{53}$

(B) $\frac{2}{21}$

(C) $\frac{1}{6}$

(D) $\frac{5}{3}$

Question 59 (JAM 2013 • Q6 • ID: JAM_2013_Q6)

MCQ

+2 M

Neg: -0.67

The value of the integral

$$\iint_D \sqrt{x^2 + y^2} dx dy, \quad D = \{(x, y) \in \mathbb{R}^2 : x \leq x^2 + y^2 \leq 2x\}$$

is

(A) 0

(B) $\frac{7}{9}$

(C) $\frac{14}{9}$

(D) $\frac{28}{9}$

Question 60 (JAM 2012 • Q4 • ID: JAM_2012_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 The value of $\int_{z=0}^1 \int_{y=0}^z \int_{x=0}^y x y^2 z^3 dx dy dz$ is

(A) $\frac{1}{90}$

(B) $\frac{1}{50}$

(C) $\frac{1}{45}$

(D) $\frac{1}{10}$

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Question 61 (JAM 2011 • Q4 • ID: JAM_2011_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 Let V be the region bounded by the planes $x=0$, $x=2$, $y=0$, $z=0$ and $y+z=1$. Then the value of the integral $\iiint_V y dx dy dz$ is

(A) $\frac{1}{2}$

(B) $\frac{4}{3}$

(C) 1

(D) $\frac{1}{3}$

Question 62 (JAM 2010 • Q2 • ID: JAM_2010_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 The value of $\iint_G \frac{\ln(x^2 + y^2)}{x^2 + y^2} dx dy$, where $G = \{(x, y) \in \mathbf{R}^2 : 1 \leq x^2 + y^2 \leq e^2\}$, is

(A) π

(B) 2π

(C) 3π

(D) 4π

Question 63 (JAM 2009 • Q8 • ID: JAM_2009_Q8)

MCQ

+6 M

Neg: -2.0

Q.8 Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and $a > 0$. Then the integral $\int_0^a \left[\int_0^x F(y) dy \right] dx$ equals

- (A) $\int_0^a y F(y) dy$ (B) $\int_0^a (a-y) F(y) dy$
 (C) $\int_0^a (y-a) F(y) dy$ (D) $\int_a^0 y F(y) dy$

Question 64 (JAM 2008 • Q7 • ID: JAM_2008_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 The set $U = \left\{ x \in \mathbb{R} \mid \sin x = \frac{1}{2} \right\}$ is

- (A) open (B) closed
 (C) both open and closed (D) neither open nor closed

Question 65 (JAM 2005 • Q12 • ID: JAM_2005_Q12)

MCQ

+6 M

Neg: -2.0

12. Let $y = f(x)$ be a smooth curve such that $0 < f(x) < K$ for all $x \in [a, b]$. Let

L = length of the curve between $x = a$ and $x = b$

A = area bounded by the curve, x -axis, and the lines $x = a$ and $x = b$

S = area of the surface generated by revolving the curve about x -axis between $x = a$ and $x = b$

Then

- (A) $2\pi KL < S < 2\pi A$
 (B) $S \leq 2\pi A < 2\pi KL$
 (C) $2\pi A \leq S < 2\pi KL$
 (D) $2\pi A < 2\pi KL < S$

Question 66 (JAM 2005 • Q10 • ID: JAM_2005_Q10)

MCQ

+6 M

Neg: -2.0

10. The integral $\int_0^1 \left[\int_0^{1-z} \left(\int_0^2 dx \right) dy \right] dz$ is equal to

(A) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^2 dx \right) dz \right] dy$

(B) $\int_0^1 \left[\int_0^{1-y} \left(\int_0^1 dx \right) dz \right] dy$

(C) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-z} dx \right) dz \right] dy$

(D) $\int_0^2 \left[\int_0^2 \left(\int_0^{1-y} dx \right) dz \right] dy$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Integral Calculus (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	4.0 to 4.0
2	NAT	+2	1.1 to 1.3
3	MCQ	+2	A
4	MCQ	+2	A
5	MCQ	+1	MTA
6	MCQ	+1	A
7	NAT	+2	MTA
8	NAT	+1	2.31 to 2.35
9	NAT	+1	2.98 to 3.02
10	MCQ	+2	B
11	MCQ	+1	A
12	NAT	+2	4.4 to 4.6
13	NAT	+2	4.70 to 4.90
14	NAT	+2	1.95 to 2.05
15	NAT	+1	4.50 to 4.60
16	NAT	+1	0.24 to 0.26
17	MCQ	+2	A
18	NAT	+2	0.49 to 0.51
19	NAT	+1	3.99 to 4.01
20	MSQ	+2	A, C
21	MCQ	+2	D
22	MCQ	+2	B

Q. No.	Type	Marks	Official Key
23	MCQ	+1	A
24	NAT	+2	4.15 to 4.17
25	NAT	+1	0.49 to 0.51
26	MCQ	+2	C
27	NAT	+1	0
28	MCQ	+1	A
29	NAT	+1	0.25 to 0.25
30	MSQ	+2	A;D
31	MCQ	+2	B
32	MCQ	+2	B
33	MCQ	+2	A
34	NAT	+2	6.60 to 6.80
35	NAT	+1	5.30 to 5.50
36	NAT	+1	2.60 to 2.70
37	MCQ	+2	B
38	MCQ	+1	C
39	NAT	+2	6.30 to 6.70
40	NAT	+2	0.230 to 0.250
41	NAT	+2	5.99 to 6.01
42	MCQ	+2	B
43	MCQ	+1	B
44	NAT	+2	2.9 to 3.1

Q. No.	Type	Marks	Official Key
45	MSQ	+2	A, B, D
46	MCQ	+2	A
47	MCQ	+1	D
48	NAT	+2	2.0 to 2.0
49	NAT	+2	0.35 to 0.4
50	MCQ	+2	MTA
51	MCQ	+2	MTA
52	MCQ	+1	C
53	NAT	+2	5.09 to 5.10
54	NAT	+2	0.125
55	NAT	+1	6.28 to 6.29
56	MCQ	+2	C
57	MCQ	+1	A
58	MCQ	+2	B
59	MCQ	+2	D
60	MCQ	+6	A
61	MCQ	+6	D
62	MCQ	+6	B
63	MCQ	+6	B
64	MCQ	+6	B
65	MCQ	+6	C
66	MCQ	+6	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.